



Isolation Bearing Design

103 Shear Stiffness

Example 04 - rivtbook

proj. 0001



Table of Contents	1
1.3 Rubber Elastic Properties	1
1.3 - 2 Shear Stiffness Example	1



1.3 | Rubber Elastic Properties

Table 1: IRHD hardness scale (MN/m²)

Hardness IRHD +/-2	Young's modulus E, MN/m ²	Shear modulus G, MN/m ²	k	Bulk modulus B(inf), MN/m ²
30	0.92	0.30	0.93	1000
35	1.18	0.37	0.89	1000
40	1.50	0.45	0.85	1000
45	1.80	0.54	0.80	1000
50	2.20	0.64	0.73	1030
55	3.25	0.81	0.64	1090
60	4.45	1.06	0.57	1150
65	5.85	1.37	0.54	1210
70	7.35	1.73	0.53	1270
75	9.40	2.22	0.52	1330

Table 2: Shore A hardness scale (lbf/in²)

Shore A (approx.)	Young's modulus lbf/in ²	Shear modulus lbf/in ²	k	Bulk modulus lbf/in ²
35	168	53	0.89	142 000
45	256	76	0.80	142 000
55	460	115	0.64	154 000
65	830	195	0.54	171 000
75	1340	317	0.52	189 000

Note

1. The majority of springs are in the hardness range 40-60 IRHD.
2. Theoretically, with a Poisson's ratio of 1/2, B should equal 3G. This is true for soft gum rubbers. For harder rubbers containing a fair proportion of non-rubber constituents, thixotropic and other effects increase 3G to about 4G.

1.3 - 2 | Shear Stiffness Example

Table 3: Bearing Specifications

variable	value	[value]	description
G_1	58.00 lb_in2	0.40 MPA	shear modulus



variable	value	[value]	description
K ₁	200000.00 lb_in2	1378.95 MPA	bulk modulus
rn _l	51	51	number of rubber layers
rdia	39.50 inch	100.33 cm	bearing diameter
rthk	0.40 inch	1.02 cm	layer thickness
sdia	38.00 inch	96.52 cm	shim diameter
sthk	0.09 inch	0.23 cm	11 guage shim thickness
cpi	3.1418	3.1418	contant pi

Bearing Stiffness for Circular Bearing

Eq. 1: rubber height

$$rht = rn_l \cdot rthk$$

$$rht = 20.40 \text{ inch} \quad [rht] = 51.82 \text{ cm} \quad | \text{ rubber height}$$

<u>rn_l</u>	<u>rthk</u>
51	0.40 inch
<u>number of rubber layers</u>	<u>layer thickness</u>

Eq. 2: bearing height

$$bht = rht + sthk \cdot (rn_l - 1)$$

$$bht = 24.95 \text{ inch} \quad [bht] = 63.37 \text{ cm} \quad | \text{ bearing height}$$

<u>sthk</u>	<u>rn_l</u>	<u>rht</u>
0.09 inch	51	20.40 inch
<u>11 guage shim thickness</u>	<u>number of rubber layers</u>	<u>rubber height</u>

Eq. 3: shear stiffness

$$K_{s1} = \frac{G_1 \cdot cpi \cdot rdia^2}{4 \cdot rht}$$

$$K_{s1} = 3.48 \text{ k}_i \quad [K_{s1}] = 6.10 \text{ kN_cm} \quad | \text{ shear stiffness}$$



G_1	rht	cpi	rdia
58.00 lb_in2	20.40 inch	3.1418	39.50 inch
shear modulus	rubber height	constant pi	bearing diameter

Shape Factor for Circular Bearing

Eq. 4: shape factor 1

$$sh_{1 \ 1} = \frac{sdia}{4 \cdot rthk}$$

$sh_{1 \ 1} = 24$ $[sh_{1 \ 1}] = 24$ | shape factor 1

sdia	rthk
38 inch	0 inch
shim diameter	layer thickness

Eq. 5: shape factor 2

$$sh_{2 \ 1} = \frac{bht}{rdia}$$

$sh_{2 \ 1} = 1$ $[sh_{2 \ 1}] = 1$ | shape factor 2

bht	rdia
25 inch	40 inch
bearing height	bearing diameter

